

PA3-2026

CS300 - CSC13002 - Introduction to Software Engineering

Project Assignment 3 (PA3-2026)

Total scores: 90 points

Duration: 2-3 weeks

This is a teamwork assignment.

A - Revised Project Plan - 2nd submission (5 points)

Based on the TA feedback from PA2-2026, revise your project plan to address the identified issues and improve its clarity and detail. Additionally, update the schedule and other sections with any new information available at this stage.

Important: Your team must include a `Changes.md` file that clearly lists all changes made compared to the PA2-2026 version. Use a single `Changes.md` file per PA submission, with separate sections for each revised document. Without this file, you will not receive points for this section.

B - Detailed Vision Document - 2nd submission (5 points)

Update and complete the vision document based on the TA feedback from PA2-2026. Detail product functional and non-functional requirements, alternatives and competitors, user environments.

Important: Document all changes in the same `Changes.md` file used for Section A. Without documenting the changes, you will not receive points for this section.

C - Use-Case Model (10 points)

Based on the requirements documented in the vision document, model use cases of the system in the form of use-case diagrams (one or several diagrams). Each diagram includes use cases, actors, and relationships among them.

Requirements:

- Use-case diagrams must be drawn using **Mermaid** format and embedded in Markdown.
- Each diagram must include: actors, use cases, and relationships (include, extend, generalization) where applicable.
- Use-case diagrams must cover all functional requirements documented in the vision document.
- Place the use-case diagrams in the use-case specification document or in a separate file.

D - Use-Case Specification (45 points)

Write a use-case specification for **every** use case in the use-case diagram. You may organize all specifications in a single Markdown file or across multiple files. Each use case must have a clear header title, and use cases should be ordered by workflow or by importance.

Each use-case specification must include:

- Use-case name and ID
- Actor(s)
- Description
- Preconditions
- Basic flow (main success scenario) - described step by step
- Alternative flows — list **all** possible alternative flows, not just the most common ones
- Postconditions
- Special requirements (if any)

Prototype Requirement: For each use case, your team must create a UI prototype using design tools (e.g., Figma, Google Stitch, v0, Bolt, Excalidraw, or similar). AI-powered tools are encouraged as they can generate prototypes directly from your use-case specifications. The prototype should show the screens involved in the use case's basic flow and alternative flows. Include screenshots of all prototype screens alongside each use-case specification.

The goal of this prototype requirement is to ensure that each use case is described clearly enough to be implementable. If you cannot generate a coherent UI from your use-case specification, the specification needs to be refined.

E - Implement 1 Functional Group using Spec Kit (20 points)

Your team must implement **1 functional group** from your project, covering the full stack: frontend, backend, and database. Your team is free to choose any popular and actively maintained tech stack (e.g., React, Vue, Flutter, Django, Express, etc.).

Requirements:

- Use **Spec Kit** to drive the implementation process: create specifications, generate plans, break down tasks, and implement following the spec-driven workflow.
- The functional group must be implemented end-to-end (UI + API/logic + data persistence).
- Your team may complete all Spec Kit steps including test generation, but you are NOT required to review or refine the generated test cases at this stage.
- Submit:
 - A video demo with narration explaining the implemented feature. Upload the video to YouTube (Unlisted or Public) and include the link in your submission.
 - The complete source code (exclude `node_modules`, `venv`, build artifacts, and other generated directories).
 - The Spec Kit artifacts generated during the process (specs, plans, tasks files).
- Ensure that dependency folders (e.g., `node_modules`) are removed before submitting.

F - AI Usage Report and Weekly Report (5 points)

AI Usage Report

Each team must follow the AI usage guidelines to declare whether AI tools were used during this sprint. If AI tools were used, a detailed log of all usages must be submitted.

Weekly Report

Conduct and document the following meetings: Sprint Planning, Scrum meetings, and Sprint Review. Follow the instructions in the "WeeklyReports" (provided separately) document.

Include Jira task screenshots showing task assignments and progress for this sprint.

Submission Guidelines

- All documents must be written in **English** using **Markdown** format. Diagrams should be drawn using **Mermaid** syntax.
- In each report/document section, the team must indicate who performed the work, who reviewed it, and who edited the section. Place this information on the first line immediately after the section header.
- Submit both the **original Markdown files (.md)** and **PDF files converted from Markdown**.
- Provide a **Git log** (screenshot or exported log) showing commit history of your repository.
- Compress all files into a directory named `PA3-Group[GroupId]` and submit as a zip file.
Example: `PA3-Group01.zip`

Grading Criteria

The assignment will be graded according to how well your team prepares the required documents and implementation. Criteria include:

- English writing: easy to understand, few errors.
- Well-formatted texts, diagrams, and charts.
- Changes are clearly documented in `Changes.md`.
- Use-case diagrams are in Mermaid format and cover key functional requirements from the vision document.

- Each use case has a complete specification with a main flow and is accompanied by prototype screenshots.
- The implemented functional group is functional end-to-end and follows the Spec Kit workflow.
- Source code is well-organized, and the video demo with narration clearly shows working features.